# WORKFORCE DECISION STUDIO

## 90-day product strategy for Medika Nusantara

A decision product for reskilling and role transition - built on a trustworthy workforce data layer.

Deliverables included

1. Product and data strategy • 2. Working prototype • 3. Synthetic datasets

# Executive position

## Do not build an attrition predictor first. Build a reskilling and role-transition decision product.

The CHRO does not need “AI workforce insights.” She needs a defensible way to decide which employee groups should be assessed, reskilled, or moved before targeted roles change - without replacing source systems and without payroll data. The 90-day PoC therefore proves two things: (1) a contextualized data layer can convert fragmented HR records into decision-grade evidence, and (2) HR managers can act through a prioritized validation queue rather than a passive dashboard.

| Assumption | Why it is explicit |
|---|---|
| Scope unit | The PoC uses a representative 200-person sample across role families and branches; production rollout would scale to all 15,000 employees. |
| Legal boundary | No Oracle payroll data, payroll-derived proxy, or compensation inference is used. |
| Outcome definition | A “successful decision” is a validated reskilling or transition action, not a model score. |
| Ground truth | Manager validation, course completion, skill assessment and mobility outcomes become the feedback loop; historical labels alone are insufficient. |
| Role exposure | Two-year role exposure is initially expert-configured and later calibrated with business process and technology-roadmap evidence. |

# 1. Problem discovery

## 1.1 Actual product problem

Product problem: HR leaders cannot reliably identify and validate employees for role-transition interventions because workforce evidence is fragmented, non-comparable, and incomplete. The job is not to generate predictions; it is to reduce the time and risk involved in moving from evidence to a reskilling decision.

- **Primary user:** Regional/branch HR manager accountable for workforce actions, supported by HR data stewards and functional managers.
- **Job to be done:** “When a role family is exposed to change, help me identify who should be assessed first, understand why, and initiate the right validation or development action.”
- **Success metric:** Validated action rate: percentage of recommendations that become an agreed assessment, learning plan, mobility conversation, or data-correction task within 30 days.
- **Guardrail:** No adverse employment action may be taken solely from a model recommendation.

## 1.2 Focus-area trade-off

| Focus area | Data availability | Business impact | 90-day feasibility | PoC speed | Decision |
|---|---|---|---|---|---|
| Attrition | Low: payroll | High, but errors | Low-medium | Medium | Defer |

| | | | | | |
|---|---|---|---|---|---|
| prediction | unavailable; historical performance unreliable | create employee-relations risk | | | |
| Skill-gap + reskilling | Medium: LMS is messy but recoverable; role/title signals available | High and directly tied to transformation | High | High | Prioritize |
| Internal mobility matching | Medium: IDs and skills must be resolved first | High; converts reskilling into placement | Medium | Medium | Design next |
| Strategic workforce planning | Medium-low: demand scenarios and role forecasts not yet validated | Very high at executive level | Low for individual decisions | Low | Use as aggregate lens, not PoC core |

Choice: skill-gap and role-transition prioritization wins because it creates an actionable output using recoverable evidence, demonstrates the value of contextualization, and establishes reusable primitives for later mobility and workforce-planning products.

## 1.3 First two weeks: critical learning agenda

| Critical question | How I would answer it |
|---|---|
| Which decision will the CHRO defend at Day 90? | Run a 90-minute decision-backwards workshop with CHRO, HRBPs, L&D and transformation lead; force a single decision, owner, cadence and success criterion. |
| Which role families are genuinely changing? | Interview operations, sales, IT and transformation leaders; map process/technology changes to 3-5 exposed role families and obtain named executive sponsors. |
| What data may legally be processed? | Complete a field-level data inventory and privacy review; classify direct identifiers, sensitive attributes, allowed purposes, retention and access roles. Confirm payroll exclusion in writing. |
| Can identities be resolved without payroll? | Profile SuccessFactors, Workable and Moodle keys; sample 500 records; test deterministic and probabilistic match rates; quantify ambiguous and unmatched cases. |
| What does “skill” mean locally? | Facilitate role-to-skill calibration with L&D and 8-12 managers; reconcile Moodle categories with a configurable taxonomy and proficiency scale. |
| Which evidence do managers trust? | Shadow 5 HR managers reviewing employee cases; test explanation formats, confidence language and required drill-down evidence. |
| What outcome can become ground truth? | Define feedback events: manager agrees/disagrees, assessment result, course completion, transition accepted, and outcome after 90/180 days. |

## 1.4 Explicit 90-day non-goals

- Individual attrition prediction or automated “flight risk” labeling: labels and features are too weak, and payroll is unavailable.
- Company-wide role obsolescence forecast for every job: initially constrain to 3-5 sponsored role families.
- Autonomous employee decisions: every recommendation is decision support and requires manager validation.
- Replacing or rebuilding HRIS/ATS/LMS: use connectors and a canonical layer above them.

- Reconstructing pre-2022 performance history: quarantine it unless independently validated.
- Production-grade MLOps for all 15,000 employees: build a thin, auditable vertical slice and a scaling design.

# 2. Data strategy

## 2.1 From fragmented records to usable evidence

**1. Contract -** Field-level source contracts, allowed purpose, owner, refresh cadence, quality tests and legal classification.

**2. Profile -** Measure completeness, uniqueness, scale drift, missingness patterns and branch-specific semantics.

**3. Resolve identity -** Create a crosswalk and canonical employee ID using deterministic rules first, then probabilistic matching.

**4. Standardize -** Normalize titles, branches, dates, rating scales and skill vocabulary; retain raw values and lineage.

**5. Enrich carefully -** Infer skills only from approved evidence such as title taxonomy, completed courses and assessments; attach provenance and confidence.

**6. Serve -** Publish AI-ready employee-role-skill snapshots plus a review queue; never overwrite systems of record.

**7. Learn -** Capture manager validation and downstream actions as new labeled evidence.

## 2.2 Logical architecture

| Layer | Responsibility | Key controls |
| --- | --- | --- |
| Sources | SuccessFactors, Workable, Moodle, custom performance tool | Read-only connectors; payroll excluded |
| Raw landing | Immutable source snapshots | Encryption, source timestamp, schema version, access logging |
| Quality + observability | Completeness, freshness, uniqueness, scale and drift tests | Fail/quarantine thresholds; branch-level scorecards |
| Identity graph | Cross-system person entities and source-ID crosswalk | Deterministic/probabilistic evidence, match score, reversible merges |
| Canonical workforce model | Employee, position, role, skill, learning, performance event | Bitemporal history, raw-to-canonical lineage |
| Contextualization services | Title normalization, skill taxonomy mapping, rating-scale normalization, inference | Model/rule version, confidence, provenance |
| Feature + decision layer | Point-in-time features and reskill recommendations | No leakage, eligibility rules, reason codes, confidence calibration |
| Experience layer | Decision queue, evidence drawer, review workflow | Role-based access, human approval, feedback capture |

## 2.3 Identity resolution without Oracle

Treat identity resolution as an evidence graph, not a one-time join. Oracle ID MN-444 is unavailable and therefore not required for the PoC. The canonical crosswalk links only approved source IDs and can later add payroll as a separately governed edge if policy changes.

- **Deterministic pass:** Exact normalized work email; approved HRIS-to-LMS mapping; exact employee number embedded in an allowed field; exact name + branch + start date when unique.
- **Probabilistic pass:** Weighted similarity on normalized name, email local-part, phone hash if allowed, branch, role family and start-date proximity. Use Fellegi-Sunter-style evidence or an explainable gradient model; do not use a black-box LLM for final linkage.
- **Decision thresholds:** ≥0.92 auto-link; 0.75-0.91 human review; <0.75 remain unlinked. Conflicting high-trust identifiers block auto-merge.
- **Merge safety:** Preserve every source record, match evidence, rule/model version and reviewer decision. Support split/undo.
- **Quality metric:** Precision on auto-links is the primary guardrail; target ≥99%. Recall can improve gradually through review.

# 3. AI system design

## 3.1 90-day component: Reskill Transition Recommender

The PoC ranks employees within an approved role family for a specific target-role pathway. It does not decide who is “good” or “bad”; it recommends who should enter a manager validation conversation first.

| Feature family | Examples | Use |
|---|---|---|
| Role exposure | Expert-configured 24-month exposure by canonical role; process automation roadmap | Creates urgency, not an individual judgment |
| Skill evidence | Observed LMS skills, completed learning, assessments, approved role-based inferred skills | Estimates target-role gap |
| Trajectory | Normalized 2022-2025 performance trend and consistency | Indicates capacity to absorb transition; missingness remains explicit |
| Engagement proxy | Allowed survey/engagement score where available | Weak supporting feature, never sole trigger |
| Mobility constraints | Branch, role eligibility, tenure bands, manager validation | Filters infeasible recommendations |
| Data quality | Identity, skill, performance and record confidence | Calibrates recommendation confidence and review routing |

Output per employee: target role, reskill-priority score, transition-readiness score, top evidence/reason codes, missing-evidence warnings, record confidence, recommendation confidence, and next-best action (manager validate, assess skills, correct data, or create learning plan).

## 3.2 Missing data strategy

- Distinguish unknown from absent. A blank skill field never means “no skill.”
- Impute only for model continuity; never display imputed values as observed facts.
- Infer candidate skills from normalized role, completed Moodle courses, assessment evidence and manager-validated equivalents. Store source, method, timestamp and confidence for every inferred skill.
- Use missingness indicators as features because branch/system adoption patterns are informative, while auditing them to avoid penalizing poorly digitized branches.
- Cap recommendation confidence when critical evidence is inferred; route low-confidence high-impact cases to review.

- Create an active-learning queue: ask managers the smallest number of questions that most improves decision confidence.

### 3.3 Confidence signaling

| Band | User sees | System behavior |
|---|---|---|
| High (≥85%) | "Ready to discuss" with 2-3 reason codes and evidence links | May enter manager queue; still requires human confirmation |
| Medium (70-84%) | "Validate evidence" plus exactly what is inferred or missing | Prompt targeted skill assessment or manager confirmation |
| Low (<70%) | "Do not act - data review required" | No recommendation; create steward task |
| Human-review override | Conflicting identity, inferred critical skill, out-of-distribution role/branch, sharp model-manager disagreement, or protected-attribute proxy concern | Block automation and log review outcome |

### 3.4 Day 75: "82% accuracy, but it feels wrong"

I would stop the launch decision. Accuracy is not sufficient evidence of usefulness and may hide class imbalance, label leakage, branch bias or a mismatched outcome definition.

1. Reconstruct the metric: define label, base rate, split strategy, time leakage and comparison baseline. Report precision/recall, calibration and error cost by branch/role - not only accuracy.

2. Run structured error review with the three managers. For each disputed case, separate data error, explanation failure, policy disagreement and legitimate model discovery.

3. Check feature and label integrity: rating-scale normalization, missingness effects, duplicate leakage, post-outcome features, and proxy variables.

4. Compare against simple baselines and manager predictions. A complex model must beat role rules and manager-only judgment on the chosen decision metric.

5. Recalibrate or reframe. If ground truth is weak, downgrade from prediction to risk triage and collect prospective validation labels.

6. Launch only as a monitored shadow workflow until agreement, calibration and intervention usefulness meet predefined gates.

# 4. Product and UX design

### 4.1 Decision surface, not dashboard

| Surface | What the HR manager needs |
|---|---|
| Transformation lens | Select exposed role family, target pathway, branch and decision deadline. |
| Prioritized queue | Employee, current→target role, priority, confidence band, reason summary and required action. |
| Evidence drawer | Observed vs inferred skills, normalized performance trend, identity quality, source lineage and missing |

| | |
|---|---|
| | evidence. |
| Action panel | Create manager validation task, request skill assessment, propose learning pathway, mark not suitable, or send to data steward. |
| Team summary | Counts of ready/validate/review, expected skill gaps and intervention capacity - no vanity charts. |
| Feedback capture | One-click agree/disagree plus reason, resulting action and later outcome. |

## 4.2 Communicating uncertainty without eroding trust

Show uncertainty as an operational next step, not as statistical decoration. Use three plain-language states: Ready to discuss, Validate evidence, and Data review required. Always pair confidence with its cause (“skills inferred from role taxonomy,” “start date conflicts,” “performance history missing”) and the action that resolves it. Avoid decimal probabilities for consequential people decisions.

# 5. 90-day roadmap: every milestone is demo-able

| Day | C-suite demo | Product learning / exit criteria |
|---|---|---|
| 0-14 | Decision contract: one exposed role family, one target pathway, named user, success metric, privacy boundary and sample evidence cards | CHRO signs the decision scope; legal confirms no payroll; managers agree on review language |
| 30 | Data Trust Console: source profile, canonical schema, title taxonomy v1, first identity graph, and before/after records for 50 employees | ≥99% precision on auto-linked validation sample; data gaps and owners are visible |
| 60 | Working decision queue for 200 employees: normalized titles, skill inference with provenance, confidence bands, and manager feedback workflow | ≥70% of sampled recommendations judged plausible enough to validate; critical errors categorized |
| 75 | Shadow evaluation: model/baseline comparison, calibration, branch/role error slices, disagreement review | No unresolved leakage; recommendation confidence calibrated; launch gate decision |
| 90 | C-suite decision pilot: one role-family transition portfolio, named interventions, review backlog, value-of-data comparison and scale plan | Managers create real actions; CHRO sees how contextualization changes decisions and approves next rollout |

# 6. From custom engagement to reusable platform

Productize the invariants; configure the client-specific semantics. The platform is not one universal model. It is a governed assembly system for workforce evidence and decision workflows.

| Reusable across clients | Configurable per client | Never silently customized |
|---|---|---|
| Connector framework and raw contracts | Source-field mappings and refresh cadence | One-off SQL embedded in production |
| Canonical employee-position-role-skill-event model | Organization/branch hierarchy and local role aliases | Client-specific tables without schema ownership |
| Identity resolution service + review UI | Match thresholds and allowed identifiers | Irreversible merges |
| Taxonomy service with versioning | Role-to-skill mappings, proficiency, language aliases | Taxonomy changes without lineage |
| Data-quality rules and scorecards | Thresholds and accountable owners | Missingness treated as absence |
| Modular AI services: normalize, infer, match, rank, explain, calibrate | Use-case policy, role exposure, eligibility, interventions | A monolithic client-trained model |
| Decision workflow and feedback event schema | Approval chain and HR operating model | Scores without action capture |
| Evaluation harness and model cards | Business metric and error-cost weights | Accuracy-only launch gates |

## 6.1 What the next five clients reuse

- 70-80% of the technical spine: connectors, canonical model, identity graph, quality tests, taxonomy tooling, confidence framework, evaluation harness and decision components.
- 40-60% of the semantic layer: common healthcare/distribution roles and skills can seed configuration, but must be locally validated.
- 100% of feedback-event definitions so every engagement compounds Rakamin's evidence about which recommendations and interventions work.
- No raw client data is pooled by default. Cross-client learning uses permissioned, de-identified patterns, reusable ontology assets and federated/aggregate insights under explicit contracts.

# 7. Synthetic data and value-of-data demonstration

Dataset A contains 220 source-like rows representing 200 employees, including 20 duplicate-ish records. 47% of primary records have blank skill fields, titles vary by branch, performance scales and history are inconsistent, IDs differ, and some dates conflict. Dataset B contains 200 canonical employees with normalized titles, resolved IDs, observed/inferred skill provenance, field confidence and human-review flags.

| Measure | Messy input | Contextualized input | Why it matters |
|---|---|---|---|
| Skill completeness | 53% | 100% populated; inferred fields labeled | Unknown is separated from absent |
| Actionable recommendations | Confidence-capped; many cases blocked | 186 high-confidence records; 6 review flags | HR receives a safe queue, not a false ranking |
| Top-40 overlap with contextualized priority | 60% | Reference decision set | Raw output wrongly flags or misses employees because aliases and blanks distort features |
| Identity | System-specific IDs + | Canonical ID + crosswalk + | Prevents duplicate |

| | | | |
|---|---|---|---|
| | duplicate-ish rows | reversible evidence | intervention and leakage |
| Explanation | Raw fields with unclear semantics | Reason codes, provenance, confidence and next action | Builds trust through inspectability |

## 7.1 Skill inference logic

For records with observed skills, values are normalized to the canonical vocabulary. For blank skill fields, Dataset B proposes three candidate skills from the validated canonical role template. In production, completed courses, assessments and manager evidence would strengthen or reject each candidate. Role-only inference is capped at 68% skill confidence and remains visibly marked; it is never represented as a verified employee capability.

# 8. Measurement, risk and governance

| Category | Metric / gate |
|---|---|
| Decision value | Validated action rate; time from exposed role selection to intervention; manager adoption |
| Recommendation quality | Precision@K for manager-validated candidates; calibration; disagreement resolution rate |
| Data layer | Identity auto-link precision; canonical completeness; unresolved conflicts; lineage coverage |
| Fairness | Selection and error-rate slices by branch, tenure band, gender/age only where legally allowed and solely for audit; never use protected attributes to rank |
| Safety | Zero automated adverse decisions; 100% low-confidence cases blocked; access and review logs complete |
| Platform | Configuration vs custom-code ratio; connector reuse; taxonomy reuse; onboarding time for next client |

# 9. Deliverable index and usage

- prototype.html - standalone browser prototype with embedded synthetic sample; toggle messy vs contextualized data and open employee decision evidence.
- medika_messy.csv / .json - source-like enterprise reality.
- medika_clean.csv / .json - canonical post-contextualization records.
- README.md - local run instructions, file descriptions, inference logic and public-deployment steps.

# 10. AI tools and working method

I used generative AI as a structured copilot to: decompose the case requirements; pressure-test scope choices; generate and validate synthetic edge cases; draft the standalone HTML/CSS/JavaScript prototype; and improve document clarity. I retained product judgment for the core decisions, explicitly constrained the synthetic-data logic, inspected generated outputs, and used deterministic scripts for reproducibility. In a real engagement, no production employee data would be entered into a public AI tool without an approved enterprise environment, data-processing agreement and privacy review.

# Least-confident decision

The decision I am least confident about is using role exposure as a meaningful 24-month signal in the first PoC. It is directionally aligned with the CEO's transformation mandate, but it can become opinion dressed as data if the process and technology roadmap is vague. I would validate it through a structured role-impact workshop with business leaders, process owners and employees, then compare expert exposure ratings with observed task changes and external role evidence. Until that validation is complete, exposure remains a configurable scenario input - not a learned fact.

# Appendix A - Dataset data dictionary

## Messy dataset

- source_row_id
- hris_employee_id
- ats_candidate_id
- lms_user_id
- employee_name
- email
- branch
- job_title_raw
- hris_start_date
- ats_start_date
- skills_raw
- performance_history_raw
- engagement_score
- record_source
- known_conflict

## Clean dataset

- canonical_employee_id
- employee_name
- branch
- hris_employee_id
- ats_candidate_id
- lms_user_id
- normalized_job_title
- target_role
- canonical_start_date
- skills
- skill_source
- inferred_skills
- performance_history
- engagement_score

- role_obsolescence_risk
- skill_gap_score
- reskill_priority_score
- transition_readiness_score
- identity_confidence
- skill_confidence
- record_confidence
- human_review_flag
- review_reason